Type MS · Calculate area, volume and surface area to volume ratio

Answer 1 of 6 · Question 3 part a.i · 3 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

Question number					

3	(a)	(i)	surface area = 24 (mm^2);
			volume = 8 (mm^3);
			3:1;
				Full marks for correct final answer	3

		(ii)	C 92;		1
		(iii)	movement from a high concentration to a low concentration;		1
		(iv)	• reduces the time taken to become colourless/rate of diffusion/reaction is faster; • the diffusion distance is shorter/greater area over which diffusion/collisions can occur;		2
	(b)	(i)	• use acid heated to different temperatures/idea of different temperatures; • compare/measure the time taken to become colourless/compare rate of diffusion;		2
		(ii)	Any two from: • the side length of the cube/size of the cube/surface area/volume of cube; • acid concentration; • concentration of alkali/indicator in the cube;	accept volume of acid Allow volume of indicator	2

Total 11 marks

Type MS · Calculate area, volume and surface area to volume ratio

Answer 1 of 6 · Question 3 part a.i · 3 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate area, volume and surface area to volume ratio

Answer 2 of 6 · Question 6 part c.i · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part c.i in this mark scheme extract

<table>
<tr><td>6 (a) (i)</td><td>Increase/goes up;</td><td>allow equivalent</td><td>1</td></tr>
<tr><td>(ii)</td><td>Information required about other factors, such as diet/exercise/alcohol consumption/age/gender/exposure to pollution/(genetic) history/smokers or non smokers;;</td><td>Allow one mark for reference to lifestyles only</td><td>2</td></tr>
<tr><td>(b)</td><td>Less mucus removed by cilia/more mucus/build up of mucus in lungs/blocks airways/bacteria not removed;
Increased risk of (lung/respiratory) infection/coughing;</td><td></td><td>1
1</td></tr>
<tr><td>(c) (i)</td><td><table><tr><td></td><td>Surface area</td><td>Volume</td><td>SA:V ratio</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td>54;</td><td>27;</td><td>2:1;</td></tr></table></td><td>Allow for SA:V ratio 6:3 ECF</td><td>3</td></tr>
<tr><td>(ii)</td><td>Smaller surface area to volume ratio (in lungs with emphysema);
Reduced/less efficient gas exchange/diffusion of oxygen/carbon dioxide;
Less oxygenated blood/less oxygen transferred (to cells);
Less (aerobic) respiration;</td><td></td><td>Max 3</td></tr>
</table>

Type MS · Calculate area, volume and surface area to volume ratio

Answer 2 of 6 · Question 6 part c.i · 3 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate area, volume and surface area to volume ratio

Answer 3 of 6 · Question 6 part a.i · 2 marks

2024 JUN · Paper 01 · find part a.i in this mark scheme extract

6 (a) (i)	3 x 3 x 6; 54 (cm^2)	full marks for correct final answer ecf	(2)
(ii)	3 x 3 x 3; 27 (cm^3)	full marks for correct final answer ecf	(2)
(iii)	54:27; 2:1	ecf for wrong values	(2)
(b)	Any five from • red blood cells contain haemoglobin; • to transport oxygen/binds to oxygen; • blood plasma; • contains dissolved substances; • named substance e.g. nutrients/hormones/CO_2/heat; • transported through blood vessels/named blood vessel;		(5)
(c)	• diffusion; • (nutrients) higher in concentration outside the cell than inside the cell/down a concentration gradient/from high to low concentration;		(2)

Total for Question 6 = 13 marks

Type MS · Calculate area, volume and surface area to volume ratio

Answer 4 of 6 · Question 6 part a.ii · 2 marks

2024 JUN · Paper 01 · find part a.ii in this mark scheme extract

6 (a) (i)	3 x 3 x 6; 54 (cm^2)	full marks for correct final answer ecf	(2)
(ii)	3 x 3 x 3; 27 (cm^3)	full marks for correct final answer ecf	(2)
(iii)	54:27; 2:1	ecf for wrong values	(2)
(b)	Any five from • red blood cells contain haemoglobin; • to transport oxygen/binds to oxygen; • blood plasma; • contains dissolved substances; • named substance e.g. nutrients/hormones/CO_2/heat; • transported through blood vessels/named blood vessel;		(5)
(c)	• diffusion; • (nutrients) higher in concentration outside the cell than inside the cell/down a concentration gradient/from high to low concentration;		(2)

Total for Question 6 = 13 marks

Type MS · Calculate area, volume and surface area to volume ratio

Answer 5 of 6 · Question 6 part a.iii · 2 marks

2024 JUN · Paper 01 · find part a.iii in this mark scheme extract

6 (a)	(i)	3 x 3 x 6; 54 (cm^2)	full marks for correct final answer ecf	(2)
	(ii)	3 x 3 x 3; 27 (cm^3)	full marks for correct final answer ecf	(2)
	(iii)	54:27; 2:1	ecf for wrong values	(2)
(b)		Any five from • red blood cells contain haemoglobin; • to transport oxygen/binds to oxygen; • blood plasma; • contains dissolved substances; • named substance e.g. nutrients/hormones/CO_2/heat; • transported through blood vessels/named blood vessel;		(5)
(c)		• diffusion; • (nutrients) higher in concentration outside the cell than inside the cell/down a concentration gradient/from high to low concentration;		(2)

Total for Question 6 = 13 marks

Type MS · Calculate area, volume and surface area to volume ratio

Answer 6 of 6 · Question 6 part b · 3 marks

2024 JUN · Paper 02 · find part b in this mark scheme extract

6 (a)	• dim room or bright room/lamp at different distances; • allow time for (pupil) to adjust; • use a camera/video to take a picture/recording of the eye/ use of ruler; • measure the width/diameter of the pupil;	Ignore change light intensity	(3)
(b)	• diameter of pupil 6- 8 mm; • substitution 3.14 x radius of pupil; • accept any value between 28-51 mm^2;	If the diameter is used instead of the radius, then 2 marks can be awarded for any value between 113-201mm^2 Any value between 28-51 scores 3 marks	(3)
(c) (i)	• circular muscles relax; • radial muscles contract; • pupil gets larger/wider /dilates/ increase in size;		(3)
(ii)	• to allow more light to enter the eye; • to see image more clearly/better;		(2)

Total for Question 6 = 11 marks